

# Predictive Modeling Versus Valuation Heuristics: A Comparative Machine-Learning Study of Residential Real-Estate Price Prediction

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**Abstract**—Residential properties are frequently priced with simple heuristics such as average price per square foot or the mean sale price of a property type. This paper asks whether supervised machine learning can outperform such heuristics on a dataset of 1,879 residential transactions (condominiums and bungalows) described by ten property attributes. Following a four-stage pipeline—business framing, data engineering, model comparison, and deployment—we benchmark Linear Regression, Random Forest, and Gradient Boosting regressors on an identical stratified 80/20 split. Random Forest achieves the lowest held-out error (MAE = \$43,795;  $R^2 = 0.834$ ), roughly halving the error of the linear baseline (MAE = \$92,220;  $R^2 = 0.459$ ). Feature-importance analysis shows that carrying costs (insurance, property tax) and sale timing dominate price formation, while raw size and room counts contribute far less than conventional heuristics assume. The selected model is deployed in an interactive Streamlit application that contrasts model predictions with two heuristic baselines and flags listings with large valuation gaps.

**Keywords**—real-estate valuation, regression, random forest, gradient boosting, feature importance, Streamlit.

## I. Introduction

Many believe that the price of house has been determined by the supply and demand in a residential transaction. Practitioners commonly have relied on two heuristics: (a) the market-average price per square foot multiplied by the listing's area, and (b) the mean sale price of comparable property types. Both are transparent in the web environment, but ignore interactions among quality, carrying costs, and market timing, producing systematic over- and under-pricing [1].

The central business question is therefore: can predictive price modeling outperform simple valuation heuristics for home prices?

We answer it with machine learning models and surfaced through an interactive Streamlit application. The pipeline design follows the Cross-Industry Standard Process for Data Mining (CRISP-DM)-tradition of separating business understanding from modeling and deployment [2].

## II. Methodology

### A. Pipeline Architecture

The system is decomposed into five our independent stages that communicate only through typed dictionaries. Stage 2 (data engineering) performs one-hot encoding, optional column dropping, and target separation; Stage 3 (modeling) owns the

train/test split and trains each model in a fully isolated function so that any estimator can be swapped without touching the rest of the pipeline; Stage 4 (deployment) consumes the Stage-3 output to persist predictions, figures, and pickled models, and to generate business insights. The design is model-count agnostic: adding a fourth or fifth estimator requires no change downstream.

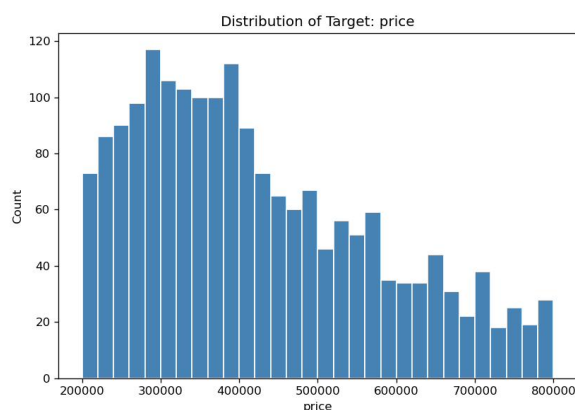
## B. Models and Evaluation

Three regression models are compared: ordinary-least-squares Linear Regression as an interpretable baseline; a Random Forest regressor [3] with 200 trees using the absolute-error split criterion; and a Gradient Boosting regressor [4] with 200 estimators, depth 3, and learning rate 0.1, all implemented in scikit-learn [5]. All models share one stratified 80/20 train/test split (random\_state = 42, stratified on the condominium indicator) to guarantee a fair comparison. Mean absolute error (MAE) on the held-out set is the primary selection criterion because it is expressed in error (dollars) and is robust to the right-skewed price distribution; RMSE and  $R^2$  are reported as secondary metrics (Fig.1, Table 1).

## III. Data processing

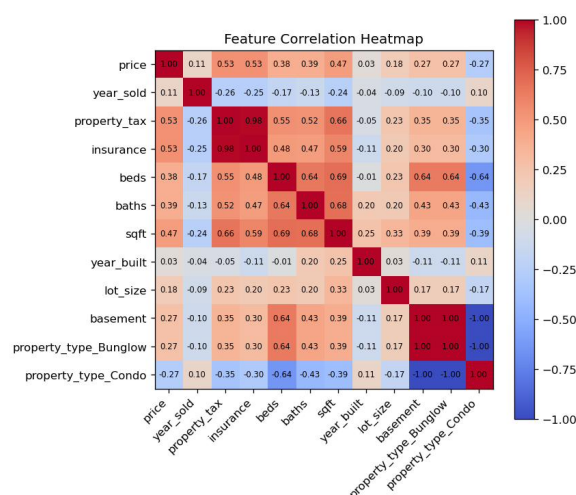
The cleaned dataset contains 1,879 residential sales and 11 columns with no missing values: sale price (target), year sold, property tax, insurance, beds, baths, square footage, year built, lot size, a basement indicator, and property type (condominium or bungalow). Prices range from \$200,000 to \$800,000 with mean \$423,055 and standard deviation \$151,538; the distribution is right-

skewed (Fig. 1). One-hot encoding of property type yields 11 model features.



**Fig. 1.** Distribution of the target variable (sale price).

Correlation analysis (Fig. 2) reveals that property tax ( $r = 0.53$ ) and insurance ( $r = 0.53$ ) correlate most strongly with price, followed by square footage ( $r = 0.47$ ), baths ( $r = 0.39$ ), and beds ( $r = 0.38$ ). Notably, property tax and insurance are nearly collinear ( $r = 0.98$ ), and the basement indicator coincides exactly with the bungalow property type in this sample.



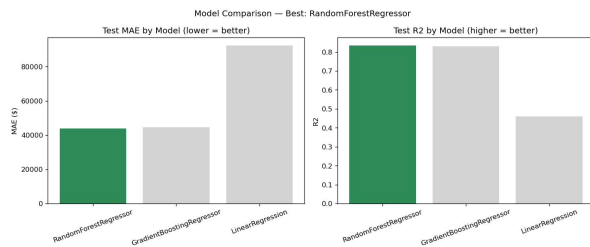
**Fig. 2.** Feature correlation heatmap including the target.

## IV. Results

**TABLE I.** Model Performance on the Held-Out Test Set

| Model             | Train MAE (\$) | Test MAE (\$) | Test RMSE (\$) | Test R <sup>2</sup> |
|-------------------|----------------|---------------|----------------|---------------------|
| Linear Regression | 91,017         | 92,220        | 116,262        | 0.459               |
| Random Forest     | 17,619         | 43,795        | 64,424         | 0.834               |
| Gradient Boosting | 33,336         | 44,611        | 65,024         | 0.831               |

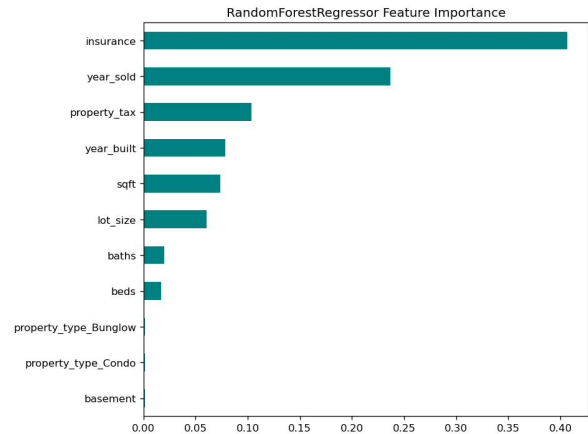
Random Forest attains the lowest test MAE (\$43,795) and highest R<sup>2</sup> (0.834), narrowly ahead of Gradient Boosting (\$44,611; 0.831) and far ahead of Linear Regression (\$92,220; 0.459, Table 1). The two ensemble methods roughly halve the linear baseline's error, indicating strongly nonlinear interactions among the price drivers (Fig. 3). Relative to the \$423K mean price, the deployed model's average error is approximately 10%.



**Fig. 3.** Test MAE and R<sup>2</sup> by model; best model highlighted.

Feature importances of the selected Random Forest (Fig. 4) rank insurance (0.41), year sold (0.24), and property tax (0.10) as the dominant drivers, followed by year built and square footage ( $\approx 0.07$  each). Property type, basement, and room counts contribute

marginally. The practical implication is that carrying costs and market timing—not raw size—carry most of the pricing signal, contradicting the assumptions embedded in \$/sqft heuristics.



**Fig. 4.** Random Forest feature importance.

For deployment, the model is served in a four-tab Streamlit application[6], in which a user specifies property attributes and receives the model prediction side by side with both heuristics (average \$/sqft and property-type average). Listings whose actual-versus-predicted gap exceeds the model's MAE are flagged for manual review, turning the model into a pricing-guideline and anomaly-screening tool.

## V. Limitations

Four caveats qualify the results. First, the Random Forest achieved a substantially lower MAE on the training data (\$17.6K) than on the test data (\$43.8K), indicating overfitting and limited generalization to unseen cases; therefore, the test results provide the more reliable measure of real-world performance. Second, property tax

and insurance show a correlation of 0.98 in the heat-map and may partly reflect the property's assessed value, making them useful proxy features for valuation review but potentially circular for pure price forecasting. Third, the model excludes another qualitative factors like neighbourhood, school, geographic, and urban-remote location features, leaving a substantial share of real-estate price variation unexplained and limiting pricing accuracy. Fourth, because the sales data span multiple years without inflation adjustment, the model represents a static market snapshot and should be redesigned to incorporate changing economic conditions and retrained as the market evolves.

## VI. Conclusion

On 1,879 residential transactions, ensemble machine learning decisively outperforms both a linear baseline and, by construction, the simple heuristics it emulates: the deployed Random Forest predicts sale price within \$43,795 on average ( $R^2 = 0.834$ ). The analysis yields a clear, actionable insight—anchor pricing guidelines on carrying costs and sale timing rather than size alone, and flag listings with large model-versus-actual gaps for review. Future work should incorporate location features, inflation-adjusted prices, and periodic retraining.

## VII. References

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